

Econ 104: Midterm review

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This note provides an overview of many of the topics from Econ 104, based on my TA notes from this quarter and past quarters. This is not intended to be exhaustive, but to provide a high-level overview of topics.

Heteroskedasticity

Problem

- Instead of $\text{var}(\epsilon_i|X) = \sigma^2$, we have $\text{var}(\epsilon_i|X) = \sigma_i^2$. For example, the plot in lecture shows the variance of errors increasing in income. Violates the identically distributed assumption of OLS.

Effects

- Given heteroskedasticity, conventional OLS standard errors are incorrect so confidence intervals and hypothesis tests that use these standard errors may be misleading.

Diagnostics

- Information visual test: after estimating a regression, you can plot the residuals against each explanatory variable and check for a pattern. If the residuals are, for example, increasing as a covariate increases, then the homoskedasticity assumption may be violated. If there is no pattern in the residuals, then homoskedasticity *may* be satisfied.
- *Breusch-Pagan* or *Lagrange Multiplier* test: uses the R^2 of a regression that describes the relationship between the squared OLS residuals and the covariates in the model. Using dependent variables x from our original regression, we run:

$$e_i^2 = \alpha_1 + \alpha_2 x_{2i} + \dots + \alpha_S x_{Si} + v_i$$

Null is $H_0 : \alpha_2 = \alpha_3 = \dots = \alpha_S = 0$. Test statistic is $NR^2 \sim \chi_{(S-1)}^2$.

- *White Test*: same as above, but in this case we use not only the covariates, but also all of their interaction terms as explanatory variables.
- *Goldfeld-Quandt*: this can be used where the sample can be partitioned into two groups and we suspect the variance could be different for the two groups, but the same within each group. The null hypothesis of the test compares the variances of the residuals of the two groups. We run our OLS regression for each group and obtain $\sigma_1^2 = SSE_1/(N_1 - K_1)$ and $\sigma_2^2 = SSE_2/(N_2 - K_2)$. Our test statistic is

$$GQ = \sigma_1^2/\sigma_2^2 \sim F_{(N_1-K_1, N_2-K_2)}$$

and our null is $H_0 : \sigma_1^2/\sigma_2^2 = 1$.

Solution

- Robust or White heteroskedasticity-consistent standard errors
- Generalized Least Squares, which accounts for the fact that the least squares estimator is no longer best. If $\sigma_i^2 = h(x_i)$, divide variables in regression by $\sqrt{h(x_i)}$.

Other topics

- None.

Autocorrelation and time series

Problem

- Instead of $cov(y_t, y_s) = 0$ for $t \neq s$, we have $cov(y_t, y_s) \neq 0 \Rightarrow cov(e_t, e_s) \neq 0$. This violates the independence of the error terms assumption of OLS.

Effects

- Confidence intervals of OLS are too narrow, but coefficients are unbiased.

Diagnostics

- ACF/PACF and Z-test. Let $k = t - s$ and $r_k \equiv cor(e_t, e_s)$. Null is $H_0 : r_k = 0, H_1 : r_k \neq 0$. Test statistic is $Z = \sqrt{t} \cdot r_k \sim N(0, 1)$. ACF/PACF will do this for you for $k = 1, 2, 3, \dots$
- Breusch-Godfrey Test. We estimate

$$e_t = \alpha_0 + \alpha_1 y_{t-1} + \dots + \alpha_k y_{t-k} + \rho_1 e_{t-1} + \dots + \rho_k e_{t-k} + v_t$$

Null hypothesis is $H_0 : \rho_1 = \rho_2 = \dots = \rho_k = 0$. Test statistic is $TR^2 \sim \chi_{(k)}^2$.

- Durbin Watson test. We estimate our test statistic $DW = \frac{\sum_{t=2}^T (e_t - e_{t-1})^2}{\sum_{t=1}^T e_t^2}$. H_0 : no autocorrelation.

Solution

- Add lags, re-run tests.
- HAC/Newey-West standard errors.

Other topics

- ARDL(p,q) models. Suppose we have a model with q lags: $y_t = \alpha + \beta_0 x_t + \beta_1 x_{t-1} + \dots + \beta_q x_{t-q} + e_t$. Immediate impact of x_t on y_t is β_0 , total effect in $t + 1$ is $\beta_0 + \beta_1$.
- Forecasting: Errors grow as period of forecast grows:
 - Let model be $y_t = \delta + \theta_1 y_{t-1} + \theta_2 y_{t-2} + \beta_1 x_{t-1} + \beta_2 x_{t-2} + e_t$
 - $\sigma_{f_1}^2 = Var(e_{t+1} | y_t, y_{t-1}) = \sigma^2$
 - $\sigma_{f_2}^2 = \sigma^2(1 + \theta_1^2)$
 - $\sigma_{f_3}^2 = \sigma^2[(\sigma_1^2 + \sigma_2^2)^2 + 1 + \theta_1^2]$
- Granger causality: $H_0 : x$ does not cause y , $H_1 : x$ does cause y . If causality runs in both directions, use VAR model. If only in one direction, use appropriate ARDL(p,q) model.

Endogeneity

Problem

- Endogeneity is when $cov(x_i, u_i) \neq 0$. When we introduced OLS, we assumed that $E[u|X] = 0$.

Effects

- Our estimate of $\hat{\beta}$ is biased. That is $\hat{\beta} = \frac{cov(x,y)}{var(x)} = \frac{cov(x,\alpha+\beta x+u)}{var(x)} = \frac{cov(x,\alpha+\beta x)}{var(x)} + \frac{cov(x,u)}{var(x)} = \beta^* + \frac{cov(x,u)}{var(x)}$, where β^* is the true value.
- Causes include **Measurement error**: if an independent variable is measured with noise, then the independent variable will be correlated with the error term.
- **Simultaneity**: when the outcome variable and independent variable are contemporaneously determined, like supply and demand.
- **Lagged-dependent variable model**: if model contains lagged values of y and the error terms in period t depend on values from $t - 1$.
- **Omitted variables**: when the residual contains an unobserved variable that is correlated with the independent variable.

Diagnostics

- Weak instruments test. We use the first stage regression

$$x_i = \alpha_0 + \alpha_1 z_1 + \dots + \alpha_p z_p + \gamma_1 x_1^c + \dots + \gamma_q x_q^c$$

where z 's are instruments and x^c are exogenous variables. Our null is $H_0 : \alpha_1 = \dots = \alpha_p = 0$. We perform an F-test. If F-statistic > 10 , instrument is strong/highly correlated with endogenous variable.

- Hausman test for endogeneity of x_i . Run 1st stage of 2SLS, get residuals $\hat{v}$. Include $\hat{v}$ as explanatory variable in original regression: $Y_i = \beta_0 + \beta_1 x_1 + \dots + \beta_j x_j + \delta \hat{v} + e_i$. Null is $H_0 : \delta = 0$. Perform t-test.
- Sargan test for exogeneity of *instruments*. Suppose we have $L > B$ instruments and B endogenous variables. We run 2SLS and get the residuals $\hat{e}_{IV}$ from the second stage. Regress $\hat{e}_{IV}$ on $z_1, \dots, z_B$. Null is that instruments are exogenous. Test statistic is $NR^2 \sim \chi^2_{L-B}$.

Solution

- Instrumental variables and 2SLS

Other topics

- Simultaneous equation models: 1) endogenous variables are those that appear in multiple equations (due to reverse causality), 2) instruments are exogenous variables in *other* equations, 3) if we have M equations, need $M - 1$ instruments to be just identified, $> M - 1$ instruments to be overidentified, and $< M - 1$ instruments to be underidentified.